

Modelling how children with Down syndrome learn language

One Bayesian toolkit, two studies: how vocabulary grows, and what reading-and-language teaching causes.



STUDY 01

Vocabulary growth

How children understand, say and sign words as they grow — across eleven international datasets.



STUDY 02

Reading & language

A causal re-analysis of a randomised trial — separating what teaching causes from what tracks ability.

One method, two studies

Both studies run on the same Bayesian foundation. What changes is the question — and how far the data can be pushed toward cause.

STUDY 01 · VOCABULARY GROWTH

Vocabulary growth

How do children build words?

Models the words children **understand, say and sign together**, pooled across eleven international studies. The comprehension–production gap falls straight out of the model as the production ratio **q**.

WHAT'S DISTINCTIVE

The understood × production-ratio decomposition; careful estimation across many datasets.

STUDY 02 · READING & LANGUAGE

Reading & language

What does teaching cause?

Re-analyses a **randomised trial** through a locked causal DAG, so the intervention's effect on reading is identified as **cause, not correlation**. Gradient-boosting discovery feeds the Bayesian estimates.

WHAT'S DISTINCTIVE

A causal DAG plus randomisation; a two-layer discovery-then-estimation method.

THE SHARED FOUNDATION

Hierarchical Bayesian models

Beta-Binomial likelihood

Gaussian-process (HSGP) trends

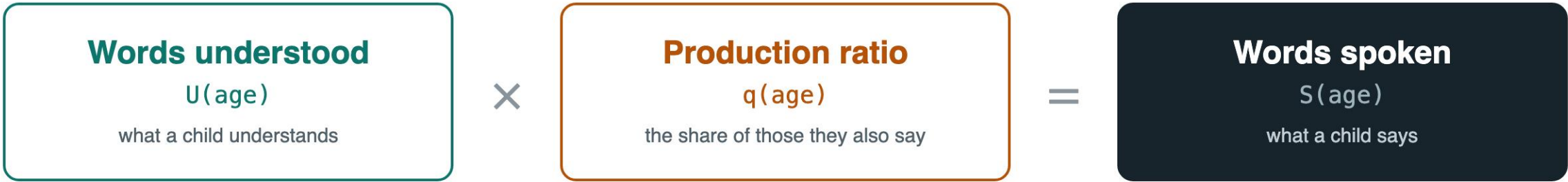
uncertainty-first – credible intervals, not p-values

reproducible: PyMC · NUTS · DuckDB · Quarto

The same engine underneath: **full-posterior inference with explicit uncertainty**, checked with predictive checks and cross-validation, and reproducible from raw data to report.

How the vocabulary model works

The modelling idea at the centre of the study: spoken vocabulary is built from what a child understands and the share of it they say.

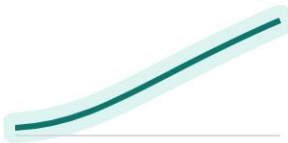


Modelling the two together means spoken vocabulary can **never exceed** what a child understands, and the comprehension–production gap falls out directly as the production ratio **q**.

THE DATA

Parent-report vocabulary checklists, **pooled across 11 international studies** onto a common 800-word scale: **588 children**, ages 8 months to about 9½ years, roughly 1,000 observations.

THE ESTIMATE

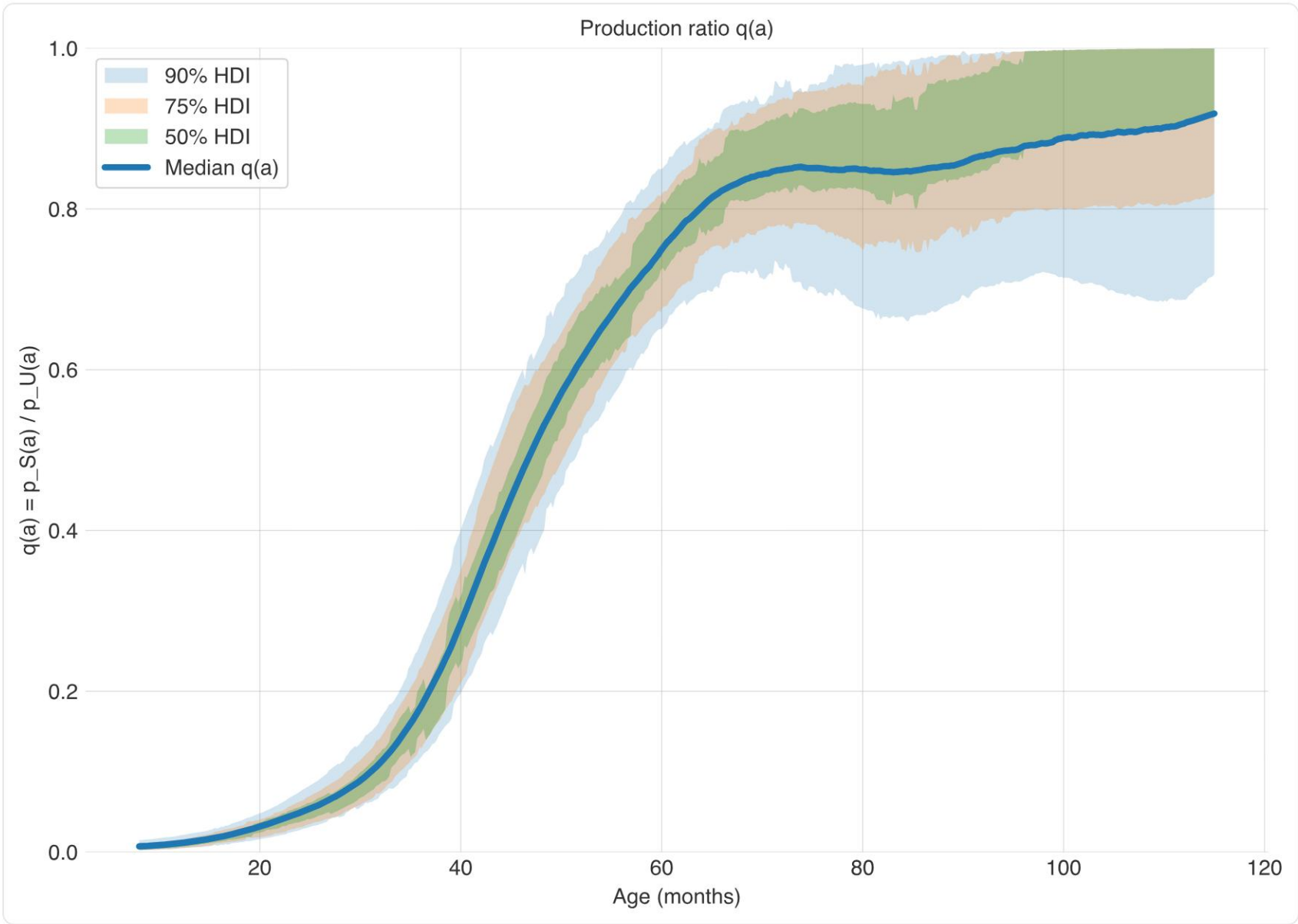


Each term is a **smooth curve over age** with full posterior uncertainty, so every result is a credible range, not a single number.

- Hierarchical Beta–Binomial
- Gaussian–process (HSGP) trends
- study & child random effects
- PyMC / NUTS
- PSIS–L00 comparison

Understanding runs years ahead of speaking

The share of understood words a child with Down syndrome also says, estimated across age from pooled international data.



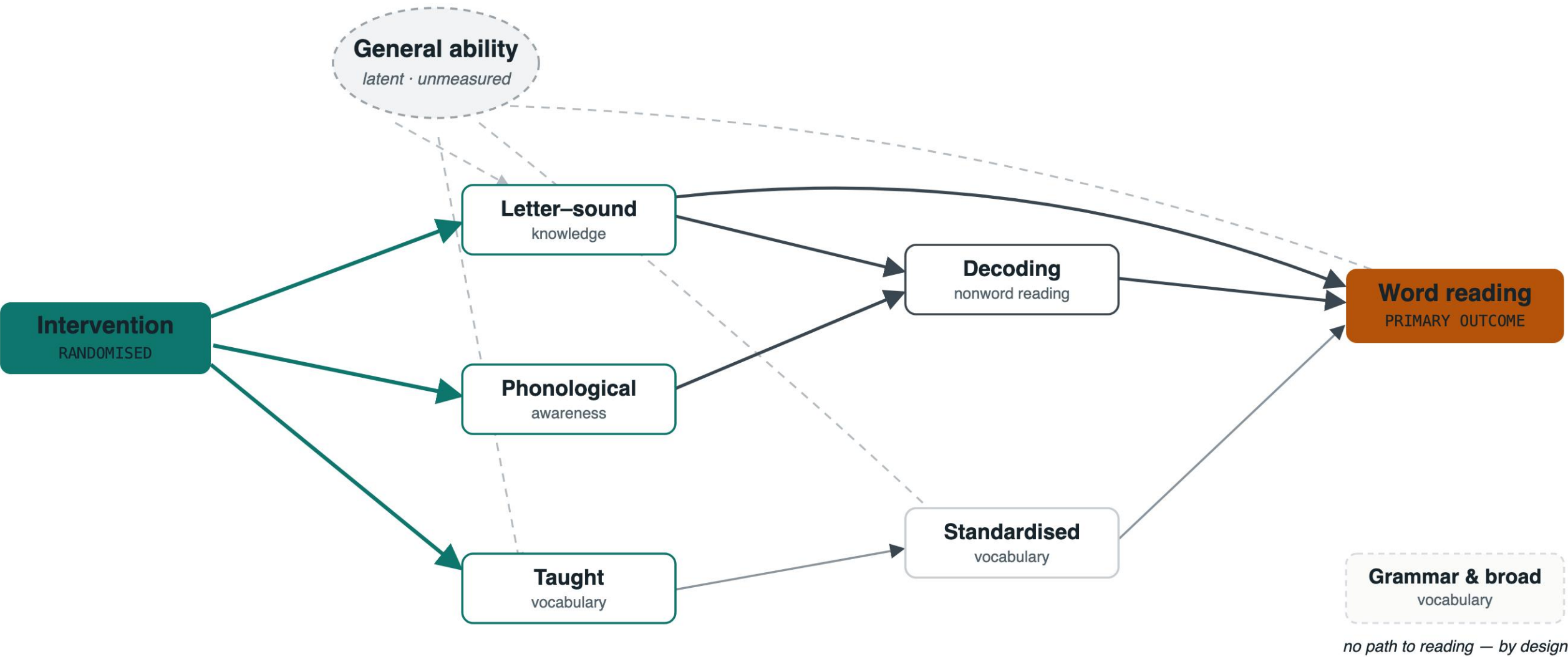
Each child's **production ratio** is the fraction of the words they understand that they also say. It climbs slowly — comprehension leads production for years.

~1% **~52%** **~86%**
at 1 year at 4 years by 7½ years

Wider than in typically-developing children at low vocabulary levels, narrowing as vocabulary grows — evidence the gap is a genuine stretch, not just a delay.

A causal design for reading progress

Separating what the teaching *causes* from what merely correlates with a child's general ability.



HOW TO READ IT

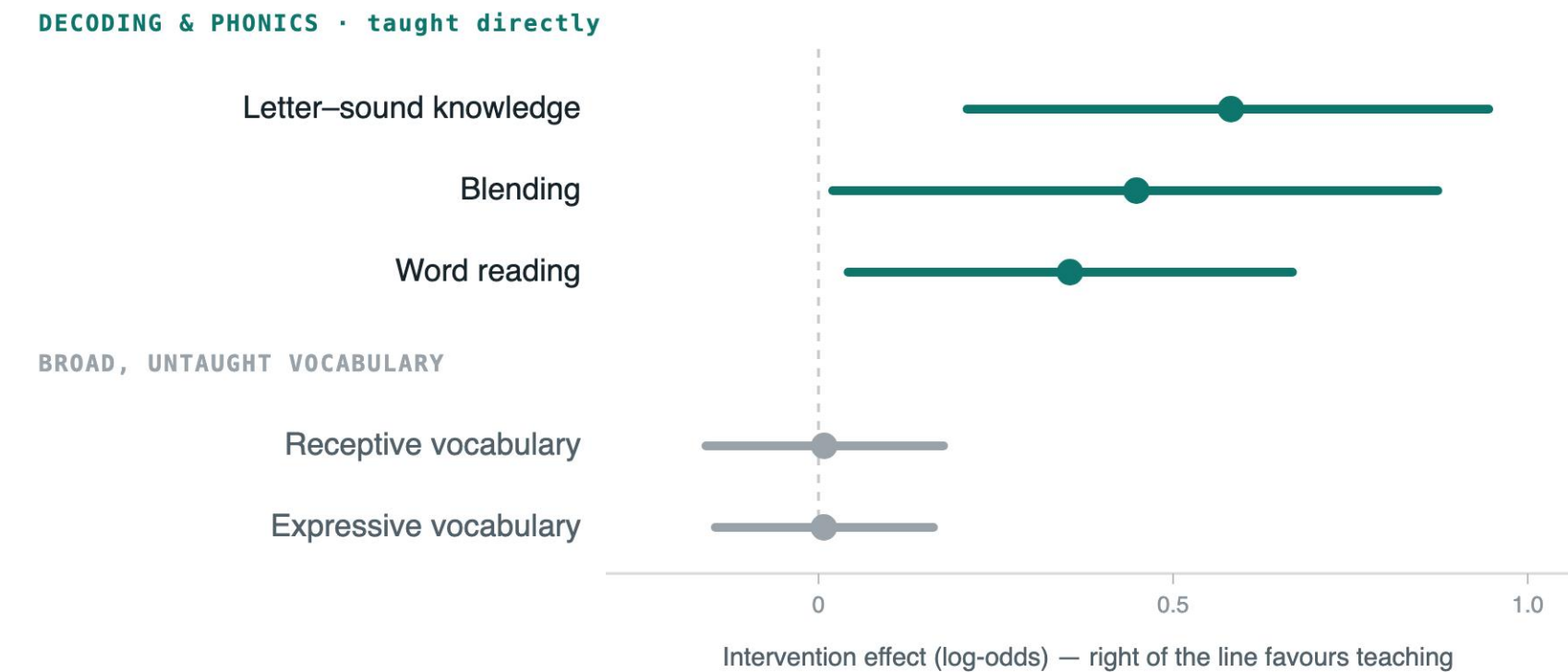
- Randomised effect — **causal**
- Hypothesised skill pathway — association
- General-ability confounding

THE IDENTIFICATION

Randomisation makes the intervention independent of general ability — so its effect on word reading is read as **genuine cause**. Every skill-to-skill link stays confounded by ability, so it is reported as **association, not cause**.

The intervention lifted reading, not broad vocabulary

The randomised effect of the teaching on each skill. Clear of the zero line means a credible benefit.



Each bar is the intervention's effect on one skill, with its **95% credible interval**. Children were randomised, so these read as cause, not correlation.

The gains cluster on the skills taught most directly: letter–sounds, blending and word reading. Broad, untaught vocabulary barely moved.

Learning transfers near what's taught, not far. The effect fades with distance from instruction. This is the transfer gradient.

● A bar clear of the dashed zero line is a credibly non-zero effect.